

Disconnected Landlords: Pilot Note

Making the foreign for-profit modal archetype estimable
(LA geography, Full-Sample, 10 July 2026)

Dmytro Kunchenko

Prepared 2026-07-10

The archetype reweighting method splits a landlord type's raw matched EPC gap (treated minus control) into a treatment effect and a composition effect. The production estimator gates which covariate cells receive their own effect on a top-100 control-pool whitelist. That whitelist is the correct base for the reweighting weights but not a valid estimability criterion, so the foreign for-profit modal cell (a 2003–2006 electric flat) is reported as not estimable. This pilot gates on matched support instead and estimates the cell directly. Sign: `current_energy_efficiency` (SAP score) higher is better, and matched effects are treated minus control, so positive means more efficient than the matched private-individual control.

1 Method

Pair difference. With 1:1 matching on exact strata, each subclass is a treated/control pair sharing all exact covariates, so the headline PSM + Subclass FE model collapses to a pair-difference regression. For pair i , $D_{y,i} = y_i^T - y_i^C$, and $D_{\text{rooms},i}, D_{\text{floor},i}$ are the within-pair differences the exact match leaves. The pooled ATT is $\hat{\tau}$ in

$$D_{y,i} = \tau + \gamma_1 D_{\text{rooms},i} + \gamma_2 D_{\text{floor},i} + \varepsilon_i,$$

clustered by local authority.

```
att <- feols(D_y ~ 1 + D_rooms + D_floor, data = est, cluster = ~local_authority)
```

Cell CATEs. Each cell $c = \text{property type} \times \text{built form} \times \text{age band} \times \text{main fuel} \times \text{floor-area tertile}$. Its effect β_c is the same regression fit separately on the pairs in c :

$$D_{y,i} = \beta_c + \gamma_{1c} D_{\text{rooms},i} + \gamma_{2c} D_{\text{floor},i} + \varepsilon_i, \quad i : c(i) = c.$$

Disjoint samples make the $\hat{\beta}_c$ independent.

```
for (c in keep_cells) {  
  m_c <- feols(D_y ~ 1 + D_rooms + D_floor,  
              data = est[cell_id == c], cluster = ~local_authority)  
  beta[c] <- coef(m_c)[1]; var_beta[c] <- vcov(m_c)[1, 1]  
}
```

Reweighting. With control-pool shares w_c renormalised over the kept cells, $\tilde{w}_c = w_c / \sum_{c'} w_{c'}$,

$$\tau_{rw} = \sum_c \tilde{w}_c \beta_c, \quad \text{Var}(\hat{\tau}_{rw}) = \sum_c \tilde{w}_c^2 \text{Var}(\hat{\beta}_c), \quad \text{composition} = \text{ATT} - \tau_{rw},$$

with coverage $= \sum_c w_c$.

```
w_norm <- w_lookup[keep_cells] / sum(w_lookup[keep_cells])  
tau_rw <- sum(w_norm * beta[keep_cells])  
tau_se <- sqrt(sum(w_norm^2 * var_beta[keep_cells]))  
composition <- att_coef - tau_rw
```

2 Matched support

A cell's matched support for treatment t is its number of intact pairs $n_{\text{pairs}}(c, t)$ and distinct local authorities $n_{\text{LA}}(c, t)$. It is distinct from the weight w_c , which is treatment-invariant: a cell rare in

the pool can be well supported for a treatment that concentrates in it. The pilot gates estimability on matched support alone:

$$\text{cell } c \text{ estimable for } t \iff n_{\text{pairs}}(c, t) \geq 50 \wedge n_{\text{LA}}(c, t) \geq 20.$$

```
cell_supp <- pd[, .(n_pairs = .N, n_las = uniqueN(local_authority)), by = cell_id]
keep_cells <- cell_supp[n_pairs >= 50 & n_las >= 20, cell_id]
```

The foreign for-profit modal cell clears the floor (over 800 pairs across 123 LAs), as do the other offshore havens; oh falls below it and reverts to the common-stock reweighting (Table 1, Figure 1).

Table 1: Estimability sweep: modal cell of every treatment under the top-100 whitelist vs the matched-support rule (EPC score)

Treat.	Modal cell (short)	Total pairs	Modal pairs	LAs	In top100	Est. 06c	Est. support	Status	τ_{rw}/cov (top100)	τ_{rw}/cov (support)
fp	Edwardian terrace	1,013,341	50,915	335	yes	yes	yes	ok	+0.20/0.58	+0.38/0.92
ukfp	Edwardian terrace	997,205	50,545	335	yes	yes	yes	ok	+0.19/0.58	+0.37/0.92
frfp	New-build electric flat	42,738	876	123	no	no	yes	ok	+0.61/0.52	+0.69/0.60
thfp	New-build electric flat	41,137	823	123	no	no	yes	ok	+0.63/0.50	+0.74/0.57
np	Post-war semi	607,446	19,061	333	yes	yes	yes	ok	+3.32/0.58	+3.15/0.89
uknp	Post-war semi	604,891	19,055	332	yes	yes	yes	ok	+3.32/0.58	+3.16/0.89
frnp	Interwar flat	3,034	27	4	no	no	no	<i>fail</i>	-0.89/0.09	-2.20/0.06
thnp	Edwardian flat	1,683	133	13	yes	yes	no	suspect [†]	-0.96/0.04	-/-
ps	Post-war semi	376,649	18,925	245	yes	yes	yes	ok	+2.66/0.58	+2.70/0.80
th	New-build electric flat	42,845	830	124	no	no	yes	ok	+0.66/0.52	+0.77/0.59
bh	New-build electric flat	39,903	827	124	no	no	yes	ok	+0.69/0.50	+0.80/0.57
eh	New-build electric flat	33,685	740	115	no	no	yes	ok	+0.84/0.47	+0.91/0.52
ch	New-build electric flat	8,776	198	39	no	no	yes	ok	+0.13/0.24	+0.23/0.23
oh	New-build electric flat	983	12	6	no	no	no	<i>fail</i>	-/-	-/-

Notes: One row per treatment = its modal (most common treated) covariate cell. ‘Est. 06c’ = estimable under the production rule (cell in the top-100 control-pool whitelist *and* ≥ 50 matched pairs); ‘Est. support’ = estimable under the pilot rule (≥ 50 matched pairs *and* ≥ 20 distinct LAs, whitelist not required). Status: **ok** = clears the support floor; suspect[†] = enough pairs but few clusters (< 20 LAs); *fail* = below the pair floor. τ_{rw}/cov = reweighted ATT and control-pool coverage under each rule. The offshore-haven 2003–2006 electric flats (frfp, thfp, th, bh, eh) flip from ‘no’ to ‘ok’ and coverage rises markedly; oh genuinely fails (too few matched pairs) – the honest “not every cell needs its own estimate”. Source: `hte_pilot_estimability_LA.csv`, `hte_pilot_summary_LA.csv` (pilot 06c).

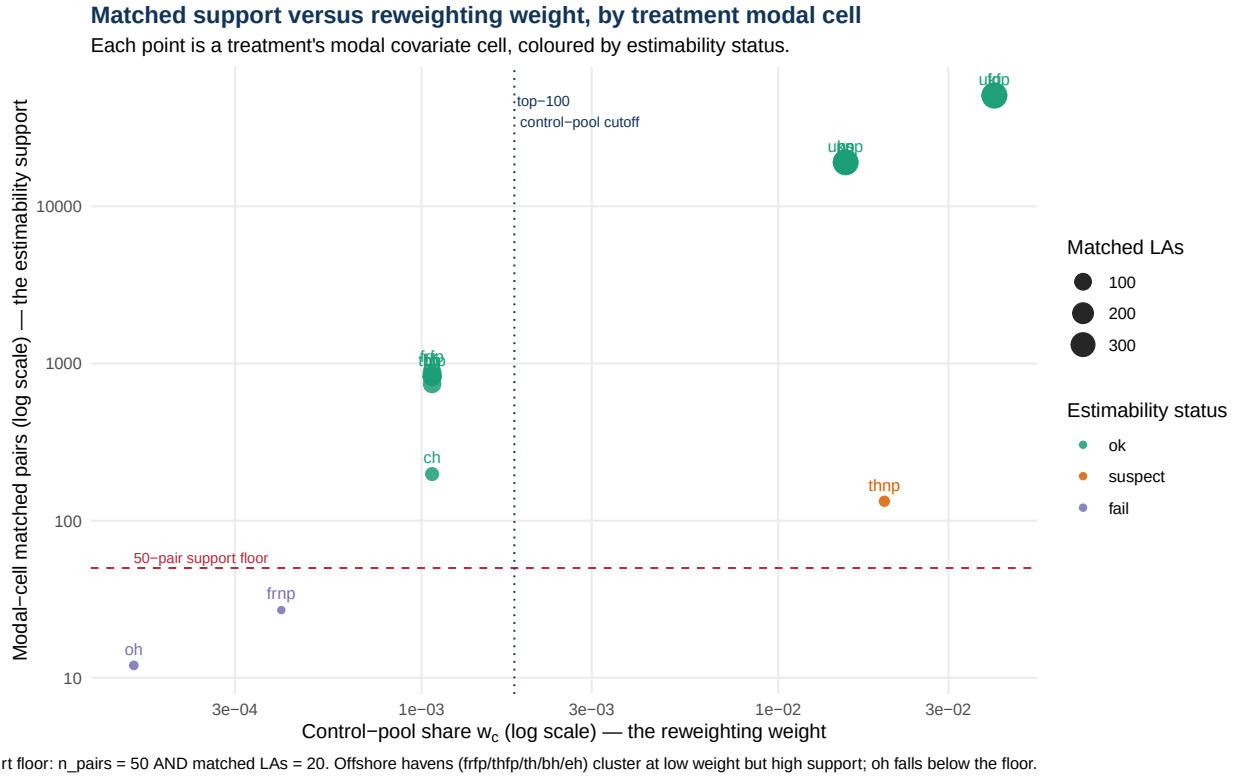


Figure 1: Matched support (vertical) against reweighting weight (horizontal) for each treatment's modal cell, coloured by estimability status. The offshore havens sit at low weight but high support; oh falls below the 50-pair floor.

3 Contribution

The reweighted effect decomposes per cell, $\tau_{rw} = \tilde{w}_{A1}\beta_{A1} + \tilde{w}_{\text{flat}}\beta_{\text{flat}} + \dots$. A1 (the modal common cell) carries a large weight but a near-zero CATE; the offshore flat carries a high CATE but a small weight. Reweighting the flat back to its small control-pool share collapses the raw foreign for-profit premium toward τ_{rw} , and the gap is the composition effect (Table 2, Figure 2).

Table 2: Two-archetype contribution to the reweighted effect (EPC score, foreign for-profit focal)

	Archetype (covariate cell)	w_c (share)	Baseline EPC	frfp β_c (SAP pts)	np β_c (SAP pts)	frfp $w_c\beta_c$ contribution
A1	Edwardian terrace	0.040	62.1	+0.23	+4.38	+0.015
Offshore flat	New-build electric flat	0.001	75.4	+2.53	+3.89	+0.005
<i>Foreign for-profit (frfp)</i> : raw ATT = +1.53; $\tau_{rw} = +0.69$ (coverage 0.60); composition = +0.83						
<i>Non-profit (np)</i> : raw ATT = +3.15; $\tau_{rw} = +3.15$; composition = +0.00						

Notes: A1 is the rank-1 (most common) cell in the eligible private-rental control pool; the offshore flat is the foreign for-profit modal cell, which the top-100 whitelist gate in the production 06c excludes but the matched-support gate makes directly estimable (over 800 matched pairs across 123 LAs). w_c = control-pool share (the reweighting weight); baseline EPC = control-pool mean score of the cell; β_c = matched-pair CATE (treated – control, LA-clustered). The contribution column uses coverage-renormalised weights and sums (with the other estimated cells) to τ_{rw} . The small weight on the high-baseline offshore flat is the source of the composition effect: reweighting to the common stock down-weights it and the raw premium collapses. Source: `hte_pilot_cells_LA.csv`, `hte_pilot_summary_LA.csv` (pilot 06c).

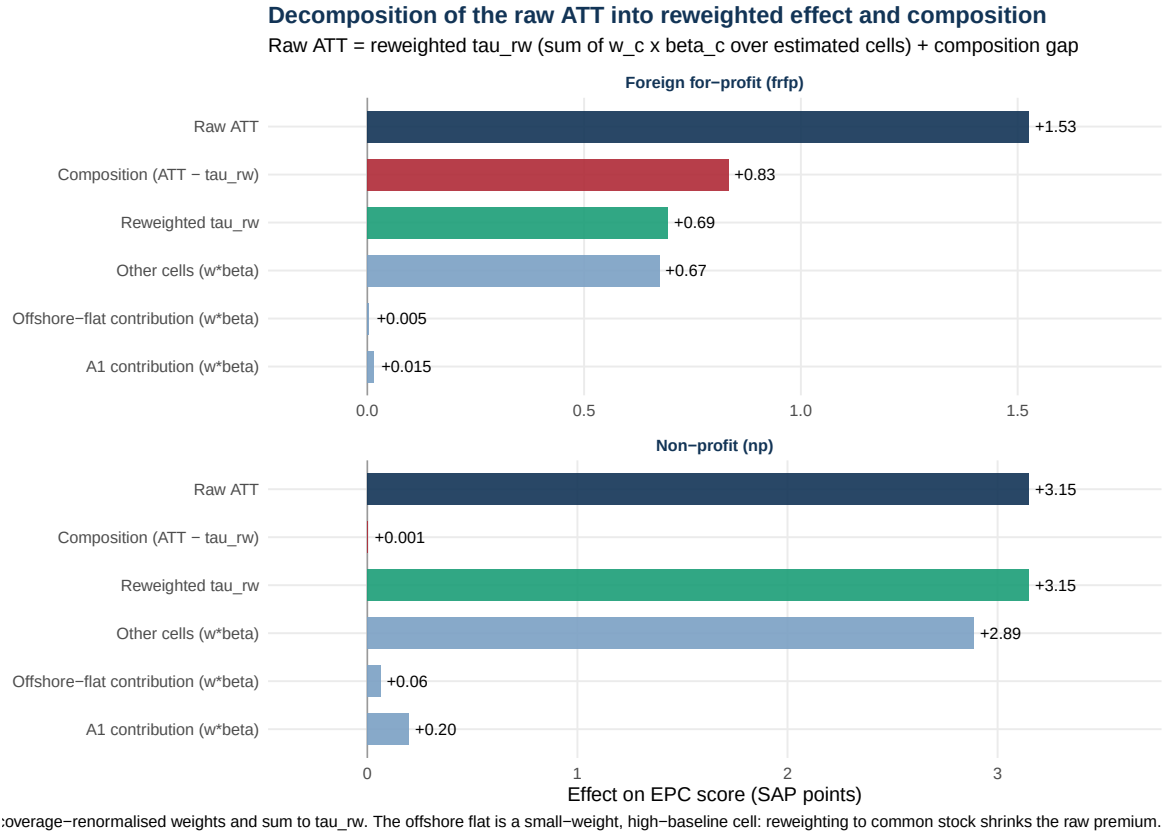


Figure 2: Decomposition of the raw ATT into $\tau_{rw} = \sum_c \tilde{w}_c \beta_c$ and the composition gap, for foreign for-profit (focal) and non-profit (contrast).